

Saayuj Deshpande

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EDUCATION

University of Pennsylvania, U.S.A

Aug 2024 – May 2026 (exp.)

Master of Science & Engineering in Robotics | GPA: 3.84/4.0

Courses: Engineering Product Management, Mechatronic System Design, Applied Machine Learning, Autonomous Vehicle Racing, State Estimation & Filtering, Reinforcement Learning, Dynamics & Control of Manipulators

Indian Institute of Technology Bombay, India

Nov 2020 – May 2024

B.Tech (Honours) in Mechanical Engineering & Minor in Controls | GPA: 8.91/10.0

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | AMP Lab, UPenn, U.S.A.

Aug 2025 – Dec 2025

Dynamic TAMP, PDDLStream, PyBullet, BiRRT, Motion Planning, Trajectory Optimization, Franka Panda Arm

- Built an adaptive local replanning framework for a PDDLStream-based manipulation system in dynamic scenes
- Implemented joint-space trajectory validation, collision checking, & BiRRT-based motion repair for a 7-DoF arm

Real-to-Simulation Research Intern | Parallax Worlds, U.S.A.

May 2025 – Jul 2025

Isaac Sim 5, Simulated Annealing, OpenVLA, Photorealistic Simulation, Mesh Processing, Standard Bots RO1

- Optimized joint stiffness & damping in Isaac Sim using annealing to minimize real-to-sim gap in joint dynamics
- Deployed OpenVLA on a 6-DoF arm & replicated the environment in Parallax Sim with reconstructed 3D meshes

Robotic Controls & Dynamics Intern | AXIBO & McMaster University, Canada

May 2023 – Jul 2023

ROS, MATLAB, Simscape, BLDC motors, Harmonic Drives, Field-Oriented Control, PID Control, Input Shaping

- Implemented FOC for BLDC motors & simulated robot links in Simscape with PID control for velocity tracking
- Applied Time-Varying Input Shaping on a 6-DoF robotic manipulator, reducing end effector oscillations by 81%

Automotive Mechatronics Intern | Mahindra & Mahindra Ltd, India

May 2022 – Jul 2022

Embedded C, CAN, Vector CANoe, CANape, In-Vehicle Networking, ECU, ABS, CAN, Unified Diagnostic Services

- Studied the automobile IVN & tested 5+ ECUs like Anti-lock Braking, Speed Control, Instrument Cluster Systems
- Performed diagnostics on CAN bus using the UDS protocol & automated 4+ diagnostic services like I/O Control

PUBLICATIONS

Drone Delivery Optimization [\[Link\]](#) | Guide: Avinash Bhardwaj, IIT Bombay

Jan 2023 – May 2024

AMPL, Gurobi, Python, Mixed-Integer Programming (MIP), MTZ Subtour Elimination, Dijkstra's Algorithm

- Optimized charging locations & generated battery-constrained shortest paths & minimal time drone trajectories

PROJECTS

Grounded-SLAM: Robust Dynamic Filtering in ORB-SLAM3

Jan 2025 – May 2025

Python, ROS2, ORB-SLAM3, Grounding DINO, YOLOv5, CNNs, Intel RealSense D435i, OpenCV, TUM RGB-D

- Combined ORB-SLAM3 with Grounding DINO for dynamic object filtering using open-vocabulary segmentation
- Evaluated the object filtering methods (YOLOv5, CNNs, DINO) to improve SLAM robustness on TUM & live data

F1Tenth Autonomous Vehicle Racing

Jan 2025 – May 2025

C++, Python, ROS2 Jazzy, Nvidia Jetson Xavier NX, VESC, Hokuyo LiDAR, RRT, SLAM, Pure Pursuit, MPC, RL

- Built a SLAM & online localization stack for autonomous cars using LiDAR & RRT* with real-time path planning
- Trained YOLO models & deployed TensorRT pipelines for real-time dynamic obstacle detection & race navigation

Pick & Place using 7-DoF Franka Emika Panda Arm

Jan 2025 – May 2025

ROS, Gazebo, RViz, Forward & Inverse Kinematics, A, Bi-RRT*, Artificial Potential Field, Bezier Interpolation*

- Created an E2E pipeline for position, velocity kinematics, motion planning & control for dynamic manipulation

Robot Wars - Mobile Robot Competition

Aug 2024 – Dec 2024

Embedded C, ATmega32U4, ESP32 C3/S2, Arduino, Time-of-Flight, HTC Vive, UDP, ESP-NOW, I2C, SPI, CAN

- Manufactured a 4-wheel autonomous robot using ESP32, L298N motor drivers & ToF sensors for wall following
- Enhanced localization & path planning with HTC Vive & enabled remote control using UDP & I2C communication